



AI Engineering Techniques & Architectures Project: **Meridian Strategy Group** **People & Travel Operations Assistant**

Business Requirements Specification

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AI Disclosure

- Generative AI tools were used during this project as part of the engineering workflow. OpenAI ChatGPT assisted with planning, drafting and review during the project's development.
- The author remains responsible for the project concept, requirements, architecture, implementation decisions, validation, evaluation and submitted content.
- AI-generated outputs informed the design and development process but did not determine the final project decisions.
- Each recommendation was assessed and revised using the project requirements, implementation evidence and observed system behaviour.
- ChatGPT assisted with defining and refining the project concept, use case, business requirements and software requirements. It also helped compare architecture and design options, identify edge cases and evaluation considerations, improve requirements traceability, explain technical concepts such as retrieval-augmented generation (RAG), agent orchestration, Model Context Protocol (MCP), deployment, CI/CD and evaluation, and review project artefacts and documentation.
- AI coding tools assisted with code generation, debugging, testing and repository-level verification. All generated code and technical recommendations were reviewed, tested and adjusted before use.
- The accompanying AI Tooling documentation identifies the tools used, their roles and the relevant AI interactions.

Executive Summary

Company Background and Business Context

Meridian Strategy Group (**MSG**) is a global management consulting organisation comprising both client-facing consulting professionals and enablement functions such as HR, Finance, IT, Legal/Risk and Operations. Client-facing employees regularly work across client engagements and locations, creating a highly mobile workforce whose time is directly connected to client delivery and revenue generation.

While on assignment, employees still need to resolve routine administrative matters such as leave, travel changes, expenses, benefits and approvals. These requests can become surprisingly complex because the correct answer may depend simultaneously on the employee's role, active assignment, travel authorisation, operational records and the applicable company policy or procedure.

For a consulting organisation, the issue is therefore not simply access to HR policies. Employees need to resolve administrative questions quickly, accurately and in the context of their current work, without unnecessarily interrupting client delivery or relying on multiple support teams and systems.

Agent Overview

Meridian Compass is a fictional internal employee assistant developed for Meridian Strategy Group (MSG). It supports employees whose administrative questions may depend on several factors at once, including their identity, current assignment or travel context, approved company policy, relevant procedures and the appropriate approval relationship.

The challenge is therefore not simply finding a policy document. Employees must also understand how the relevant rule applies to their current circumstances and determine the correct next step. Meridian Compass addresses this need by combining approved internal guidance with authorised operational information to provide a clear, employee-specific response supported by cited evidence.

Authority Model

Authority	Role
Enterprise Identity	Establishes who the user is.
Structured operational data	Establishes what situation exists.
Policy	Establishes what is permitted, required or prohibited.
Procedure	Establishes how the supported process is carried out.

General model knowledge, internet search, other companies' practices and user assertions are not authoritative MSG evidence.

Business Requirements

ID	Requirement
BR-01	Authenticate/simulate Enterprise Identity before employee-specific processing.
BR-02	Resolve self-service context from authenticated identity, not a typed employee ID.
BR-03	Retrieve approved policy/procedure evidence for material company claims.
BR-04	Access structured operational facts only through approved MCP operations.
BR-05	Support multiple multi-step people/travel workflows.
BR-06	Protect another employee's private information.
BR-07	Cite relevant internal document ID/title/section/snippet.
BR-08	Expose a concise architectural tool/source trace, not hidden chain-of-thought.
BR-09	Require explicit confirmation before consequential mock actions.
BR-10	Never perform real bookings, HR changes or financial actions.
BR-11	Fail safely when evidence/data is insufficient.
BR-12	Run reproducibly locally, in Docker, CI and hosted deployment.
BR-13	Evaluate grounding, citations, tool behaviour, safety and latency.
BR-14	Keep controlled fixtures immutable and mock-action state transient/resettable.

Primary Workflows

- International-assignment PTO.
- Role-aware business travel entitlement.
- PTO plus personal travel extension.
- Expense/per-diem reconciliation.

Business Success Criteria

- Real MCP client/server protocol boundary.
- Grounded RAG evidence from the approved corpus.
- Privacy controls based on authenticated identity.
- Citations and safe architectural traces.
- Controlled evaluation that does not invent missing company rules.

Requirement-to-outcome view

Business outcome	Requirements Enabling it
Employee-specific guidance	BR-01, BR-02, BR-04, BR-05
Grounded company guidance	BR-03, BR-07, BR-11
Privacy and safe actions	BR-06, BR-09, BR-10, BR-14
Transparency and auditability	BR-07, BR-08, BR-13
Engineering portability	BR-12, BR-13

Final Implementation Evidence

The final implementation satisfies the intended architecture with:

- a deployed web application,
- RAG over 19 controlled documents,
- 13 MCP tools,
- server-authoritative simulated identity,
- deterministic verification and a controlled 30-case evaluation.

The final evaluation result is 27/30 (90%). The remaining three failures were bounded test/contract mismatches rather than cases where the system invented missing authority.